

ON CENTRAL VERSUS FACTORIAL MOMENTS.

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Summary: For many probability distributions, especially in the case of discrete distributions, it is relatively easy to find factorial moments. Relations exist (see Comtet, 1974, p. 160) between factorial and ordinary moments. Using such relations, ordinary moments are found first customarily in order to have central moments. Needless to mention, central moments play a key role in statistical inference. In this paper, we introduce two numbers with their properties and establish, using these numbers, direct relations between factorial and central moments. We illustrate our results for modified power series distributions.

1. INTRODUCTION

Consider a discrete random variable (r.v.) X having a probability function (p.f.) $p(x, \theta)$ for $x \in N$, $\theta \in \Theta$, where N is a subset of integers and Θ a parameter space. Needless to mention, central moments $\mu_k(\lambda)$, where

$$\mu_k(\lambda) = \sum_x (x-\lambda)^k p(x, \theta) ;$$

$$\lambda = \sum_x xp(x, \theta) ;$$

play a key role in statistical inference. Finding such central moments at times involves computational difficulties. However,

it is relatively easy to find factorial moments, $\mu'_{(m)}$. Relations exist between factorial and ordinary moments, μ'_m (see Comtet, 1974, p. 160; Jordan, 1965). They are

$$(1) \quad \mu'_{(m)} = \sum_{i \geq 0}^m F(m, i) \mu'_i$$

$$(2) \quad \mu'_m = \sum_{i \geq 0}^m S(m, i) \mu'_i,$$

where $F(m, i)$ and $S(m, i)$ are called Stirling numbers of the first and second kind, respectively. These integer-valued numbers are generated by the following identities:

$$(3) \quad \frac{(\ell n(1+t))^n}{n!} = \sum_{m \geq n}^{\infty} F(m, n) \frac{t^m}{m!}, \quad \text{for } n \geq 0$$

and

$$(4) \quad \frac{(e^t - 1)^n}{n!} = \sum_{m \geq n}^{\infty} S(m, n) \frac{t^m}{m!}, \quad \text{for } n \geq 0.$$

Using (2), ordinary moments customarily are found first, and then they are converted into central moments.

Of interest to us in this paper is how to obtain central moments directly from factorial moments avoiding the intermediate step of finding ordinary moments. To accomplish this, we make use of two numbers which are introduced with their properties in Section 2. In Section 3, direct relations between central and factorial moments are established and are also illustrated for modified power series distributions. Our results confirm Riordan (1937) and Gupta and Singh (1981) as particular cases.

2. TWO NUMBERS

For a real-valued λ and nonnegative integers k and n such that $k \geq n$, we define two numbers $F(k, n, \lambda)$ and $S(k, n, \lambda)$ as given below. Both $F(k, n, \lambda)$ and $S(k, n, \lambda)$ are zero for $k < n$.

$$(5) \quad D_{\lambda} F(k, n, \lambda) = (n+1)F(k, n+1, \lambda)$$

and

$$(6) \quad D_{\lambda} S(k, n, \lambda) = -kS(k-1, n, \lambda),$$

where D_{λ} indicates the derivative with respect to λ . Characteristics of such numbers could be identified from their exponential generating functions (e.g.f.). Let ϕ_1 and ϕ_2 be their e.g.f. That is, for $|t| < 1$,

$$\phi_1 \equiv \phi_1(n, \lambda, t) = \sum_{k \geq n} F(k, n, \lambda) \frac{t^k}{k!}$$

and for $t > 0$,

$$\phi_2 \equiv \phi_2(n, \lambda, t) = \sum_{k \geq n} S(k, n, \lambda) \frac{t^k}{k!}.$$

To obtain closed forms of ϕ_1 and ϕ_2 we proceed as follows. Due to definitions (5) and (6), we obtain

$$(7) \quad D_{\lambda} \phi_1(n, \lambda, t) = (n+1)\phi_1(n+1, \lambda, t)$$

and

$$(8) \quad D_{\lambda} \phi_2(n, \lambda, t) = -t\phi_2(n, \lambda, t)$$

To solve differential-difference equations (7) and (8), we

make use of the known z-transformation technique. Let,

$$(9) \quad G(\lambda, t, z) = \sum_{n \geq 0} \phi_1(n, \lambda, t) z^n.$$

Then, it is easy to see that

$$\begin{aligned} D_z G(\lambda, t, z) &= \sum_{n \geq 0} n \phi_1(n, \lambda, t) z^{n-1} \\ &= D_\lambda G(\lambda, t, z), \end{aligned}$$

which is satisfied by

$$(10) \quad G(\lambda, t, z) = (1+t)^{\lambda+z}.$$

The reason for choosing the solution in (10) is that it leads to a function in (11) which in turn yields Stirling numbers for $\lambda = 0$. Substituting (10) into (9) and evaluating at $z = 0$, the n-th derivative with respect to z of the identity, we obtain

$$(11) \quad \phi_1 = (1+t)^\lambda \frac{(\ln(1+t))^n}{n!}.$$

Proceeding in a similar manner, we get

$$(12) \quad \phi_2 = e^{-\lambda t} \frac{(e^t - 1)^n}{n!}.$$

Comparisons of (11) and (12) with (3) and (4), respectively, reveal that when $\lambda = 0$, numbers $F(k, n, \lambda)$ and $S(k, n, \lambda)$ are nothing but Stirling numbers. While Stirling numbers are always integers, our numbers are either integers or fractions, depending upon λ . For example,

$$F(k, 0, \lambda) = \Gamma(\lambda+1)/\Gamma(\lambda-k+1)$$

$$F(n, n, \lambda) = 1$$

and

$$S(k, 0, \lambda) = (-\lambda)^k$$

$$S(n, n, \lambda) = 1.$$

Substituting expressions (5) and (6), respectively, in the derivatives of (11) and (12) with respect to λ and comparing the coefficients of $t^k/k!$, we obtain the following recurrence relations which are then used to tabulate them for a specified λ (see Tables I and II).

$$\begin{aligned} F(k+1, n, \lambda) &= \lambda F(k, n, \lambda-1) + F(k, n-1, \lambda-1) \\ (13) \quad F(k+1, n, \lambda) &= (\lambda-k)F(k, n, \lambda) + F(k, n-1, \lambda) \\ S(k+1, n, \lambda) &= -\lambda S(k, n, \lambda) + S(k, n-1, \lambda-1) \\ S(k+1, n, \lambda) &= (n-\lambda)S(k, n, \lambda) + S(k, n-1, \lambda). \end{aligned}$$

Using (11) and (3), we may write, due to the definition that $F(k, n, \lambda) = 0$ for $k < n$,

$$\begin{aligned} \sum_{k \geq 0} \sum_{n \geq 0} F(k, n, \lambda) \frac{t^k}{k!} u^n &= \sum_{n \geq 0} \sum_{k \geq n} F(k, n, \lambda) \frac{t^k}{k!} u^n \\ (14) \quad &= (1+t)^{\lambda+u} \\ &= (1+t)^\lambda \sum_{j \geq 0} \sum_{n \geq 0} F(j, n) \frac{t^j u^n}{j!} \\ &= \sum_{i \geq 0} \sum_{j \geq 0} \sum_{n \geq 0} \frac{\Gamma(\lambda+1)}{\Gamma(\lambda-i+1)} \frac{F(j, n)}{i! j!} t^{i+j} u^n \\ (15) \quad &= \sum_{k \geq 0} \sum_{n \geq 0} \sum_{j \geq n} \frac{\Gamma(\lambda+1)}{\Gamma(\lambda-k+j+1)} \binom{k}{j} \\ &\quad F(j, n) \frac{t^k u^n}{k!} \end{aligned}$$

TABLE I
F(k, n, λ)

k/n	0	1	2	3	4
0	1				
1	λ	1			
2	λ(λ-1)	λ+(λ-1)	1		
3	λ(λ-1)(λ-2)	λ(λ-1)+λ(λ-2) +(λ-1)(λ-2)	λ+(λ-1) +(λ-2)	1	
4	λ(λ-1)(λ-2)(λ-3)	λ(λ-1)(λ-2) +λ(λ-1)(λ-3) +λ(λ-2)(λ-3) +(λ-1)(λ-2)(λ-3)	λ(λ-1)+λ(λ-2) +λ(λ-3) +(λ-1)(λ-2) +(λ-1)(λ-3) +(λ-2)(λ-3)	λ+(λ-1) +(λ-2) +(λ-3)	1

TABLE II
S(k, n, λ)

k/n	0	1	2	3	4
0	1				
1	-λ	1			
2	λ ²	(1-λ)-λ	1		
3	-λ ³	(1-λ) ² -λ(1-λ) +λ ²	(2-λ)+(1-λ)-λ	1	
4	λ ⁴	(1-λ) ³ -λ(1-λ) ² +λ ² (1-λ)-λ ³	(2-λ) ² +(1-λ) ² +λ ² +(2-λ)(1-λ) -λ(1-λ)-λ(2-λ)	(3-λ)+(2-λ) +(1-λ)-λ	1

Comparing coefficients of $\frac{t^k u^n}{k!}$ on both sides of (15), we obtain

$$F(k, n, \lambda) = \sum_{j \geq n}^k \frac{\Gamma(\lambda+1)}{\Gamma(\lambda-k+j+1)} \binom{k}{j} F(j, n).$$

This implies that $F(k, n, \lambda)$ is a linear combination of Stirling numbers of the first kind. $F(k, n)$ is known (see Comtet, 1974,

p. 293) to be extremely huge as k is large. Consequently, $F(k, n, \lambda)$ is large. An asymptotic expansion for $F(k, n, \lambda)$, as k increases, is

$$(16) \quad F(k, n, \lambda) \approx \frac{\Gamma(k-\lambda)}{\Gamma(n-\lambda)}.$$

We now return to $S(k, n, \lambda)$. Using (12) and (4), we write, due to the definition that $S(k, n, \lambda) = 0$ for $k < n$,

$$\begin{aligned} & \sum_{k=0}^{\infty} \sum_{n=0}^k S(k, n, \lambda) \frac{t^k u^n}{k!} \\ &= \sum_{n=0}^{\infty} \sum_{k=n}^{\infty} S(k, n, \lambda) \frac{t^k u^n}{k!} \\ (17) \quad &= \exp[-\lambda t + u(e^t - 1)] \\ &= e^{-\lambda t} \sum_{j=0}^{\infty} \sum_{n=0}^j S(j, n) \frac{t^j u^n}{j!} \\ &= \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \sum_{n=0}^j \frac{(-\lambda)^i S(j, n) t^{i+j} u^n}{i! j!} \\ (18) \quad &= \sum_{k=0}^{\infty} \sum_{n=0}^k \sum_{j=0}^k (-\lambda)^{k-j} \binom{k}{j} \\ & \quad S(j, n) \frac{t^k u^n}{k!}. \end{aligned}$$

Comparing coefficients of $\frac{t^k u^n}{k!}$ on both sides of (18), we obtain $S(k, n, \lambda)$ as a linear combination of Stirling numbers of the second kind, namely

$$(19) \quad S(k, n, \lambda) = \sum_{j=n}^k \binom{k}{j} (-\lambda)^{k-j} S(j, n).$$

For large k , the Stirling number $S(k, n)$ is also known to be extremely large, and hence an asymptotic expansion for $S(k, n, \lambda)$, as k increases, is desirable. Using an expansion (see Jordan, 1965) for $S(j, n)$, we rewrite (19) as

$$\begin{aligned}
 S(k, n, \lambda) &= \sum_{j \geq n}^k \sum_{i \geq 0}^n \binom{k}{j} (-\lambda)^{k-j} \frac{(-1)^n \binom{n}{i} (-1)^i i^j}{n!} \\
 &= (-1)^n \sum_{i \geq 0}^n (i-\lambda)^k \frac{\binom{n}{i} (-1)^i}{n!}
 \end{aligned}$$

Hence,

$$(20) \quad S(k, n, \lambda) \approx \frac{(n-\lambda)^k}{n!}, \text{ for large } k.$$

Also, replacing t by $\ell n(1+v)$ in (12), we note that

$$\frac{v^n}{n!} = \sum_{k \geq n}^{\infty} S(k, n, \lambda) (1+v)^{\lambda} \frac{(\ell n(1+v))^k}{k!}.$$

Due to (11), we may rewrite this as

$$\frac{v^n}{n!} = \sum_{k \geq n}^{\infty} \sum_{j \geq k}^{\infty} S(k, n, \lambda) F(j, k, \lambda) \frac{v^j}{j!}.$$

Comparing similar coefficients on both sides, we then notice that

$$(21) \quad \sum_{k \geq n}^j F(j, k, \lambda) S(k, n, \lambda) = \begin{cases} 1 & \text{if } j = n \\ 0 & \text{if } j \neq n \end{cases}$$

Changing u into $(x-\lambda)$ in (14), and differentiating k times the identity with respect to t , we obtain for $t=0$,

$$(22) \quad x(x-1)\dots(x-k+1) = \sum_{n \geq 0}^k F(k, n, \lambda) (x-\lambda)^n.$$

Making use of the orthogonal property stated in (21), we obtain a complementary relation as given below:

$$\begin{aligned}
 e^{t(x-\lambda)} &= \sum_{k=0}^{\infty} (x-\lambda)^k \frac{t^k}{k!} \\
 &= \sum_{k=0}^{\infty} \sum_{j \geq 0} \sum_{n \geq j}^k S(k, n, \lambda) F(n, j, \lambda) (x-\lambda)^j \frac{t^k}{k!} \\
 &= \sum_{k=0}^{\infty} \sum_{n \geq 0}^k S(k, n, \lambda) \sum_{j=0}^n F(n, j, \lambda) (x-\lambda)^j \frac{t^k}{k!} \\
 (23) \quad &= \sum_{k=0}^{\infty} \sum_{n \geq 0}^k S(k, n, \lambda) x(x-1) \dots (x-n+1) \frac{t^k}{k!}
 \end{aligned}$$

Evaluating at $t = 0$ the k -th derivative with respect to t of the identity in (23), we obtain

$$(24) \quad (x-\lambda)^k = \sum_{n \geq 0}^k S(k, n, \lambda) x(x-1) \dots (x-n+1).$$

Many known properties of Stirling numbers may be seen as a particular case of our results with $\lambda = 0$.

In the next section we explore some statistical applications of our numbers.

3. STATISTICAL APPLICATIONS

When the p.f. $p(x, \theta)$ is

$$p(x, \theta) = P(X=x) = \frac{a_x (g_\theta)^x}{f_\theta},$$

where $a_x > 0$ for $x \in \mathbb{N}$, g_θ and f_θ are positive finite and differentiable, the r.v. X is said to have (see Gupta, 1974) a modified power series distribution (m.p.s.d.). In the case in which g_θ is invertible, we obtain a class called generalized power series distributions.

Its central moment generating function, assuming it exists, is given by

$$M_x(\lambda, t) = E[e^{t(X-\lambda)}] \\ = \sum_x \sum_{i \geq 0} \frac{a_x(g_\theta) x^i e^{-\lambda t} (e^t - 1)^{x-i}}{f_\theta}$$

Here λ is the mean of the probability distribution (in the previous section λ was any real number). Using (12), we rewrite it as

$$M_x(\lambda, t) = \sum_{k \geq 0} \sum_{i \geq 0} \sum_{j \geq 0} \frac{(j+i)!}{i!} \\ a_{j+i}(g_\theta)^{j+i} \frac{S(k, j, \lambda) t^k}{k! f_\theta}$$

Since $\mu_k(\lambda)$ is the coefficient of $\frac{t^k}{k!}$, we note that

$$(25) \quad \mu_k(\lambda) = \sum_{i \geq 0} \sum_{j \geq 0} \frac{(j+i)!}{i!} \\ \frac{a_{j+i}(g_\theta)^{j+i}}{f_\theta} S(k, j, \lambda).$$

We now obtain direct relations between central and factorial moments. These relations are useful to obtain one from the other. Note from (22) that

$$(26) \quad \mu'_k = \sum_{n \geq 0} F(k, n, \lambda) \mu_n(\lambda)$$

Similarly, from (24)

$$(27) \quad \mu_k(\lambda) = \sum_{n \geq 0} S(k, n, \lambda) \mu'_n.$$

Substituting (25) into (26), we get

$$\mu'_k = \sum_{i \geq 0} \sum_{s \geq 0} \sum_{n \geq s} F(k, n, \lambda) S(n, s, \lambda) \\ \frac{(s+i)! a_{s+i}(g_\theta)^{s+i}}{i! f_\theta}.$$

Due to the orthogonality stated in (21), this reduces to

$$(28) \quad \mu'_k = \sum_{i=0}^{\infty} \frac{(k+i)! a_{k+i}(g_\theta)^{k+i}}{i! f_\theta},$$

confirming Gupta and Singh's (1981) result. Also, by differentiating (25) and using a recurrence relation given in (13), we obtain

$$\mu'_{k+1}(\lambda) = [D_\theta \mu'_k(\lambda) + k \mu'_{k-1}(\lambda) D_\theta \lambda] / D_\theta \ln g_\theta,$$

which agrees with Gupta's (1974) result. In a similar way, it is easy to show that

$$\mu'_{(k+1)} = \frac{D_\theta \mu'_k}{D_\theta \ln g_\theta} + (\mu'_k - k) \mu'_k.$$

Now, we illustrate our results for some specific modified power series distributions.

3.1 Generalized Poisson distribution

It is given by

$$p(x, \theta) = \frac{\beta_1(\beta_1 + \beta_2 x)^{x-1}}{x!} \frac{(\theta e^{-\beta_2 \theta})^x}{e^{\beta_1 \theta}},$$

$x = 0, 1, 2, \dots$; $\theta > 0$; $\beta_1 > 0$; $\beta_2 > 0$. While $f_\theta = e^{\beta_1 \theta}$, $g_\theta = \theta e^{-\beta_2 \theta}$, $a_x = \beta_1(\beta_1 + x\beta_2)^{x-1}/x!$, using (25) we obtain that

$$(29) \quad \mu'_k(\lambda) = \sum_{n=0}^k \sum_{i=0}^{\infty} \frac{\beta_1(\beta_1 + \beta_2(n+i))^{n+i-1}}{i! e^{\beta_1 \theta}} \cdot (\theta e^{-\beta_2 \theta})^{n+i} S(k, n, \lambda),$$

where

$$\lambda = \frac{\beta_1 \theta}{1 - \beta_2 \theta}.$$

Case 1: For $\beta_2 = 0$, $\beta_1 = 1$, X_1 has a Poisson distribution. In this case $\lambda = \theta$, and from (29)

$$\mu_k(\lambda) = \sum_{n \geq 0} S(k, n, \lambda) \theta^n,$$

confirming Riordan (1937) as a special case.

Case 2: For $\beta_2 = 1$, the p.f. $p(x, \theta)$ is called a Borel-Tanner distribution (see Haight and Brenner, 1960) which is used to study a queuing phenomenon. Substituting $\beta_2 = 1$ into (29), we obtain the appropriate results.

3.2 Generalized negative binomial distribution

When the p.f. $p(x, \theta)$ is as given below, it is called a generalized negative binomial distribution.

$$p(x, \theta) = \frac{\gamma \Gamma(\gamma + \beta x)}{x! \Gamma(\gamma + \beta x - x + 1)} \frac{(\theta(1-\theta)^{\beta-1})^x}{(1-\theta)^{-\gamma}},$$

$x = 0, 1, 2, \dots$; $0 < \theta < 1$, $|\theta\beta| < 1$. While $f_\theta = (1-\theta)^{-\gamma}$,

$g_\theta = \theta(1-\theta)^{\beta-1}$, and $a_x = \frac{\gamma \Gamma(\gamma + \beta x)}{x! \Gamma(\gamma + \beta x - x + 1)}$, we obtain from

(25) that

$$\begin{aligned} \mu_k(\lambda) &= \sum_{n \geq 0} \sum_{i \geq 0} \frac{\gamma \Gamma(\gamma + \beta(n+i))}{\Gamma(\gamma + (\beta-1)(n+i) + 1)} \\ &\quad \cdot \frac{(\theta(1-\theta)^{\beta-1})^{n+i}}{i! (1-\theta)^{-\gamma}} S(k, n, \lambda), \end{aligned}$$

where

$$\lambda = \left(\frac{\gamma\theta}{1-\theta\beta} \right).$$

Case 1: For $\beta = 0$, it reduces to a binomial distribution. In this case $\gamma = \gamma\theta$, and

$$\mu_k(\lambda) = \sum_{n=0}^k S(k, n, \lambda) \frac{\Gamma(\gamma+1)}{\Gamma(\gamma-n+1)} \theta^n,$$

which includes Riordan as a particular case.

Case 2: For $\beta = 1$, it reduces to a negative binomial distribution. In which case,

$$\mu_k(\lambda) = \sum_{n=0}^k \frac{\Gamma(\gamma+n)}{\Gamma(\gamma)} \left(\frac{\theta}{1-\theta}\right)^n S(k, n, \lambda),$$

where

$$\lambda = \frac{\gamma\theta}{1-\theta}.$$

3.3 Generalized logarithmic series distribution

Consider

$$p(x, \theta) = \frac{\Gamma(x\beta)}{x\Gamma(x)\Gamma(x\beta-x+1)} \frac{(\theta(1-\theta))^{\beta-1}x}{(-\ln(1-\theta))},$$

$x = 1, 2, 3, \dots$; $0 < \theta < 1$, $\beta \geq 1$, $0 < \theta\beta < 1$. While

$$f_\theta = -\ln(1-\theta), \quad g_\theta = \theta(1-\theta)^{\beta-1}, \quad a_x = \frac{\Gamma(x\beta)}{x\Gamma(x)\Gamma(x\beta-x+1)},$$

the p.f. $p(x, \theta)$ is called a generalized logarithmic series distribution. Note that

$$\mu_k(\lambda) = \sum_{n=0}^k \sum_{i=0}^{\infty} \frac{\Gamma((n+i)\beta)(\theta(1-\theta))^{\beta-1}n+i}{\Gamma((n+i)(\beta-1)+1)i!(-\ln(1-\theta))} \\ \cdot S(k, n, \lambda),$$

where

$$\lambda = - \frac{\theta}{(1-\theta)\ln(1-\theta)}.$$

Case 1: For $\beta = 1$, the p.f. $p(x, \theta)$ reduces to a logarithmic series distribution, and

$$\mu_k(\lambda) = - \sum_{n \geq 0} (n-1)! \left(\frac{\theta}{1-\theta} \right)^n S(k, n, \lambda) / \ln(1-\theta),$$

where

$$\lambda = - \frac{\theta}{(1-\theta) \ln(1-\theta)}.$$

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